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Fundamentals of Analytics

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Spotify Wrapped (Kruthik's Version)

Music has long been an integral part of the influence on mood, feelings, and everyday life, including my own. So, when asked to complete an analytics project, I examined my Spotify streaming data compiled over the past four years. I was interested in arguments regarding the emotional effect of music, i.e., whether "sad" music would detrimentally affect one's mental condition. Hence, my analysis focused on pinpointing patterns for skipping songs according to song themes and their energetic attributes.

Data Collection and Preparation

I obtained the data directly from Spotify by making an official request for the data, which took around five days to fulfill. The main issue that I had to deal with was the treatment of the JSON format that Spotify provides, which is not ideal for direct analysis. To bypass this, I built a data pipeline that processed the JSON files and transformed them into CSV format, which is conducive to analysis. The data for each year came in separately, so I had to merge them into a single dataset that reflected my listening history. The cleaning process included renaming of columns for simplicity, deletion of unnecessary or redundant data, and exclusion of non-music entries such as podcasts and audiobooks. The cleaned and formatted dataset was now ready for exploratory analysis.

Exploratory Data Analysis

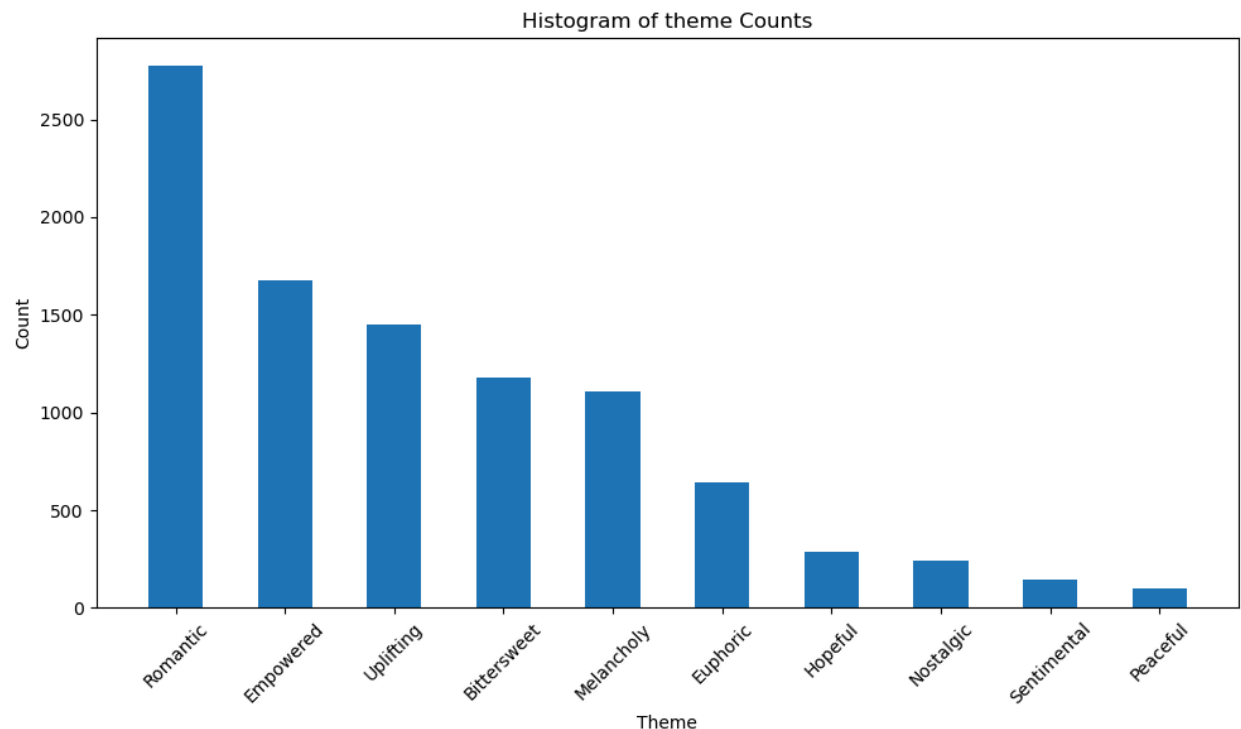
My initial analysis showed me my top 50 most-played tracks in the previous four years. It gave me some interesting insights into how my musical preferences had changed over this period. Surprisingly, my most played track was one that had not been added to my playlists, but I put it back on upon learning of its importance when I did my analysis.

Then, I considered the thematic impacts on my listening time, discovering clear correlations between long listening times and certain emotional states, particularly when I was in states of heightened happiness or sadness.

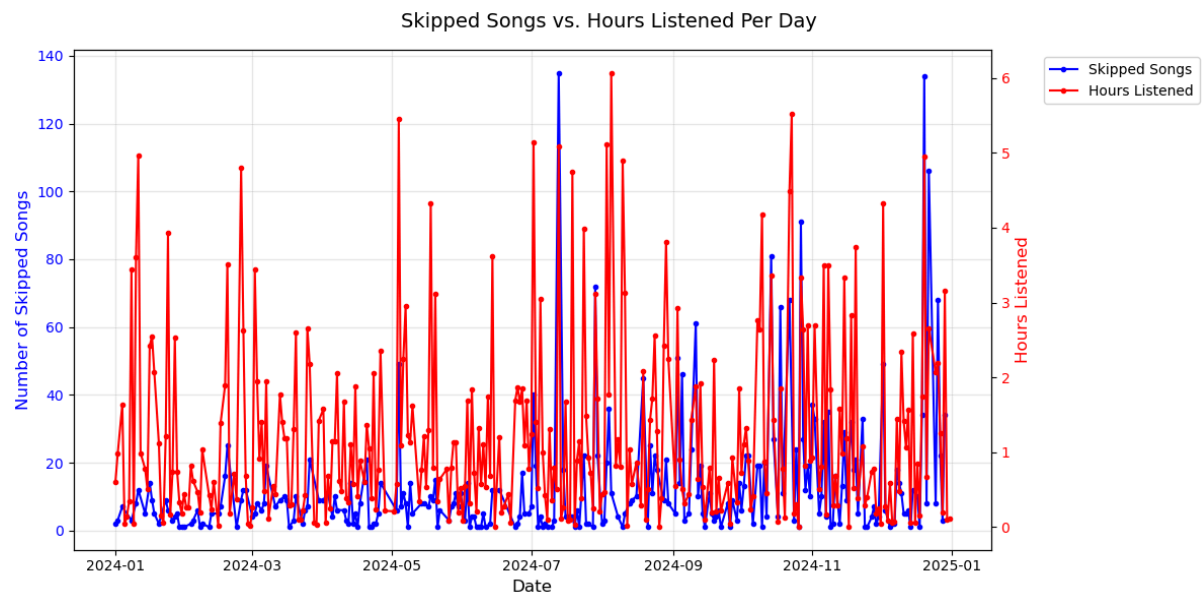
In addition to enhancing the analysis, I employed an OpenAI API to include energy ratings and thematic labels on every song. This was a computationally intensive and expensive process, and so I have limited this elaborate augmentation to the year 2024 alone. Although I later discovered Exportify, a service providing richer information about songs, I have chosen to keep my API-based data to ensure consistency in the study. I am glad I got some first-hand experience augmenting data using LLMs.

Thematic and Behavioral Insights

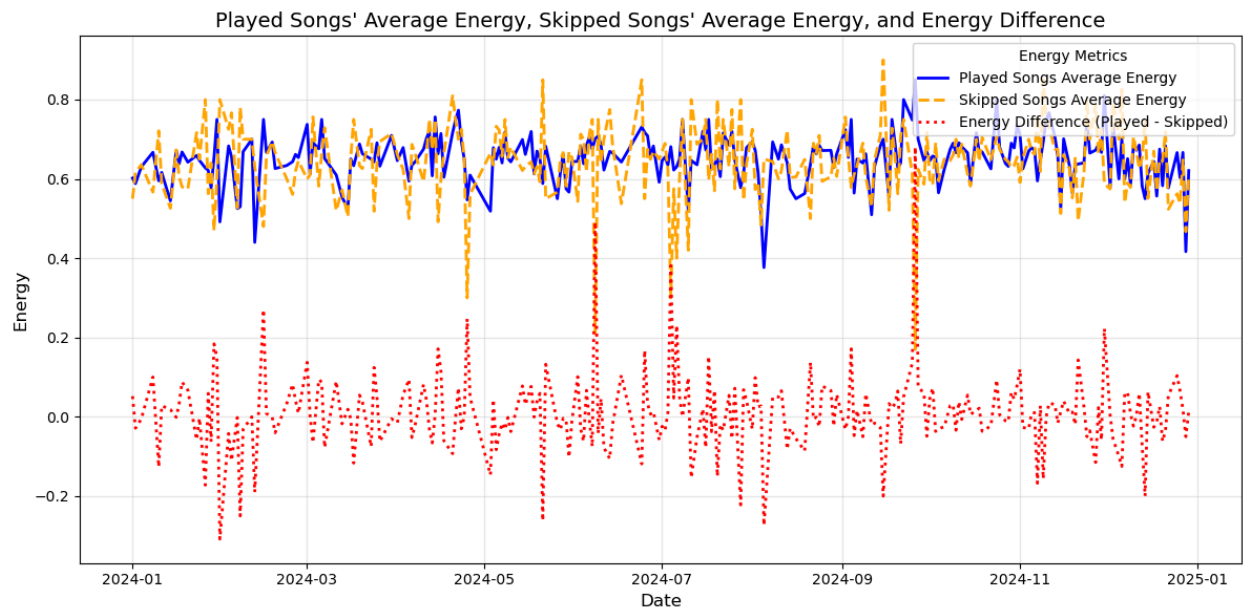
My thematic analysis demonstrated a dominant preference for romantic music, particularly Indian songs associated with heartbreak, and the least frequent themes were calm.



Initially, visualizations of typical day-to-day song-skipping patterns using romantic songs became too intricate. Simplifying these visualizations, it was clear that days with minimal song-skipping strongly correlated with my most productive days (based on personal experience), suggesting a strong association between long periods without interruptions in listening and productivity.



Analyzing song energies provided further insights into my skipped songs and the song energies linked to my listening behavior.



Predictive Modeling and Methodology

To predict song-skipping behavior, I first employed a Random Forest classifier with an initial accuracy of 67%. Hoping for better performance, especially concerning part of the day, I created a cyclical time feature via a sine transformation. This brought hourly variations into a more normalized form and enhanced the consistency of the data.

Classification Report:					
		precision	recall	f1-score	supp
ort					
	False	0.72	0.81	0.76	1
868					
	True	0.54	0.42	0.47	1
011					
	accuracy			0.67	2
879					
	macro avg	0.63	0.61	0.62	2
879					
	weighted avg	0.66	0.67	0.66	2
879					
Accuracy: 0.67					
Feature: is_english, Importance: 0.02					
Feature: daily_avg_energy, Importance: 0.48					
Feature: energy, Importance: 0.21					
Feature: hour, Importance: 0.17					
Feature: theme_Empowered, Importance: 0.02					
Feature: theme_Euphoric, Importance: 0.01					
Feature: theme_Hopeful, Importance: 0.01					
Feature: theme_Melancholy, Importance: 0.01					
Feature: theme_Nostalgic, Importance: 0.01					
Feature: theme_Peaceful, Importance: 0.00					
Feature: theme_Romantic, Importance: 0.02					
Feature: theme_Sentimental, Importance: 0.01					
Feature: theme_Uplifting, Importance: 0.01					
Feature: part_of_day_evening, Importance: 0.01					
Feature: part_of_day_morning, Importance: 0.01					
Feature: part_of_day_night, Importance: 0.01					

```
# Convert 'hour' into cyclic features
dfAn['sin_hour'] = np.sin(2 * np.pi * dfAn['hour'] / 24)
```

However the model also revealed a large discrepancy between skipped and fully played songs that impacted model recall and accuracy. I used the XGBoost algorithm to respond to this because it can effectively work with imbalanced data to yield improved prediction outcomes.

```
scale_pos_weight=y_train.value_counts()[0] / y_train.value_counts()[1]
```

Classification Report:

	precision	recall	f1-score	support
False	0.82	0.73	0.77	1868
True	0.58	0.71	0.64	1011
accuracy			0.72	2879
macro avg	0.70	0.72	0.71	2879
weighted avg	0.74	0.72	0.73	2879

Accuracy: 0.72

Feature: is_english, Importance: 0.28

Feature: daily_avg_energy, Importance: 0.43

Feature: sin_hour, Importance: 0.29

These results were still not optimal, but the optimal prediction results were obtained by combining Random Forest and XGBoost using a Voting Classifier. The ensemble leveraged the advantage of both algorithms, where Random Forest guaranteed stability through averaging predictions while XGBoost treated the data imbalance.

```

# Define Random Forest Model
rf_model = RandomForestClassifier(
    n_estimators=300,
    max_depth=10,
    class_weight='balanced', # Still balancing based on class weight
    random_state=4286
)

# Define XGBoost Model
xgb_model = xgb.XGBClassifier(
    objective='binary:logistic',
    eval_metric='logloss',
    use_label_encoder=False,
    scale_pos_weight=y_train.value_counts()[0] / y_train.value_counts()[1], # Balance class
    n_estimators=300,
    learning_rate=0.05,
    max_depth=6,
    subsample=0.8,
    colsample_bytree=0.8,
    random_state=4276
)

# Create an Ensemble Model using Voting Classifier
ensemble_model = VotingClassifier(
    estimators=[('rf', rf_model), ('xgb', xgb_model)],
    voting='soft' # 'soft' uses probability averaging for better performance
)

# Train the ensemble model
ensemble_model.fit(X_train, y_train)

# Predict on the test set
y_pred = ensemble_model.predict(X_test)

# Evaluate the model
print("Classification Report:")
print(classification_report(y_test, y_pred))
print(f"Accuracy: {accuracy_score(y_test, y_pred):.2f}")

# Feature Importance (Optional)
print("\nFeature Importance from Random Forest:")
rf_model.fit(X_train, y_train) # Fit RF separately to get importance
for feature, importance in zip(X.columns, rf_model.feature_importances_):
    print(f"Feature: {feature}, Importance: {importance:.2f}")

```

Classification Report:

	precision	recall	f1-score	support
False	0.98	0.95	0.97	3166
True	0.91	0.97	0.94	1631
accuracy			0.96	4797
macro avg	0.95	0.96	0.95	4797
weighted avg	0.96	0.96	0.96	4797

Accuracy: 0.96

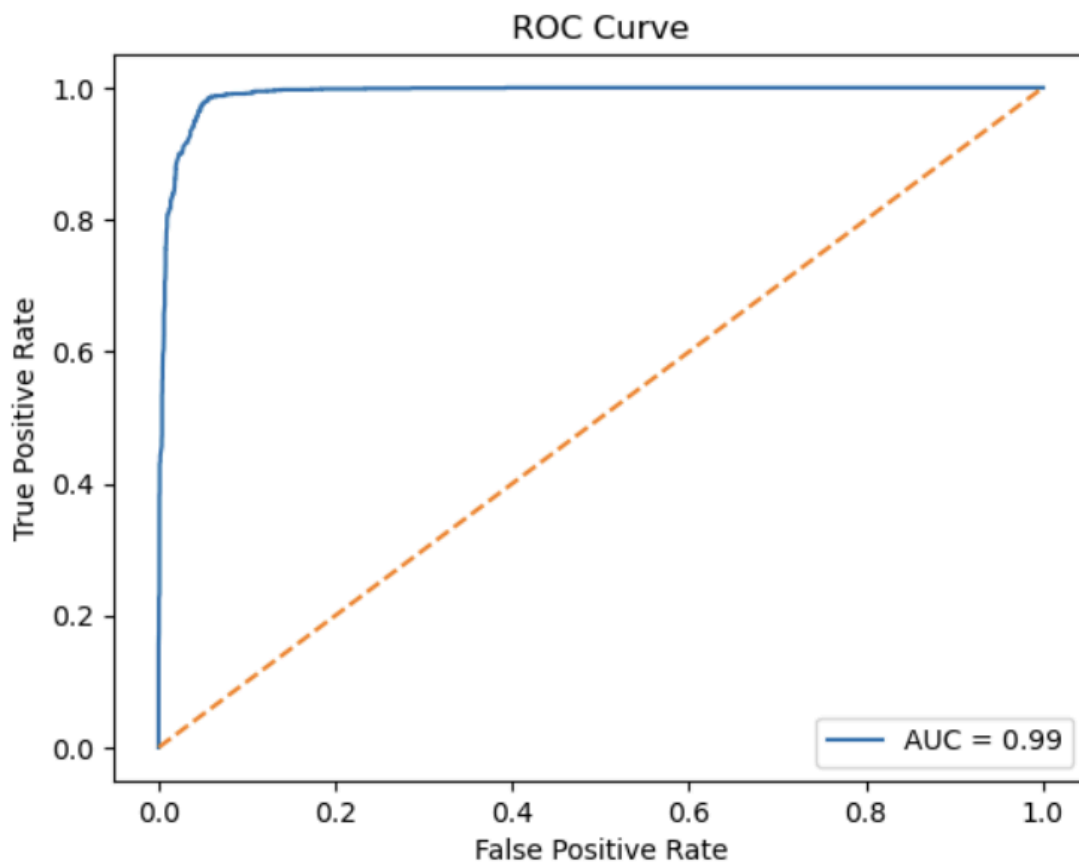
Feature Importance from Random Forest:

Feature: is_english, Importance: 0.02
 Feature: daily_avg_energy, Importance: 0.89
 Feature: sin_hour, Importance: 0.05
 Feature: platform_ios, Importance: 0.03
 Feature: platform_not_applicable, Importance: 0.01
 Feature: platform_osx, Importance: 0.02

Model Validation

I rigorously validated the predictive model using cross-validation, wherein subsets of data were tested randomly to assess the robustness of the model. ROC curve analysis also offered extensive performance insights, affirming the reliability and efficacy of the ensemble method.

Cross-Validation Accuracy: 0.9543 ± 0.0055



Conclusion

This project demonstrated that machine learning algorithms could predict song-skipping behavior accurately using the time of day and energetic qualities of songs. Not only did the results further my insight into individual listening patterns, but also illuminated the intricate dynamic between musical taste, mood, and productivity.